ReMindAR: An Adaptive Augmented Reality-Based Memory Companion for Dementia Care Using Deep Face Recognition

Kaviyaraj R [1,*]
Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankulathur, India.
https://orcid.org/0000-0002-1858-1582

Pawishrajhen A R[2]
Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankulathur, India.
pawishrajhen@gmail.com

Harish S[3]
Department of Computational Intelligence, SRM Institute of Science and Technology, Kattankulathur, India.
harishs121203@gmail.com

***Abstract*—Dementia often impairs recognition of familiar people and retrieval of contextual information during social interaction. Existing assistive systems typically provide static reminders, which can either overwhelm the user or provide insufficient support when recognition is weak. ReMindAR addresses this limitation through an adaptive augmented reality (AR) pipeline that combines MTCNN-based face detection and alignment, InceptionResnetV1 facial embeddings pretrained on VGGFace2, cosine-similarity identity matching, and an interpretable Decision Tree classifier that predicts one of four reminder intensity levels from six encounter features. The reminder model is trained on a cognitively motivated synthetic dataset and achieved 82.33% accuracy on a held-out test set. The complete system is implemented as a progressive web application supporting AR overlays and speech-to-text conversation logging, while all recognition and decision-making stages run locally on the device to preserve privacy. These results indicate that combining deep face recognition with transparent context-aware prediction is a promising direction for memory support in dementia care.**

***Index Terms*—Augmented reality, dementia care, face recognition, adaptive reminder systems, deep learning, assistive technology**

I. Introduction

Dementia is a neurodegenerative disorder that progresses over time, and millions of people are affected by this disorder worldwide. The number of dementia patients is expected to rise in the coming years [1]. The most common and frustrating problem encountered by dementia patients is the inability to identify familiar faces and the inability to recall the context while communicating with other people. Although many technologies have been developed to help dementia patients, the success of these technologies depends on their adaptability to the cognitive state of the user [2]. A recent survey on integrating augmented reality with AI in education reported improved interactive and personalized learning, indicating the broader value of adaptive AR interfaces in user-centered support settings [3].

Generative AI is transforming multiple domains through automated content creation and intelligent decision-making, reinforcing the growing role of intelligent assistance in human-centered systems [4]. Most existing reminder systems provide fixed information regardless of the level of recognition confidence. In real-world applications, this often leads to overwhelming the user with too much information when it is not needed or, conversely, not providing enough support when recognition is low [5]. This issue prompted us to investigate an adaptive approach that adjusts reminder intensity according to quantifiable recognition confidence and contextual information.

An AR framework combined with deep learning has also been shown to enhance immersive and effective learning environments, suggesting that tightly coupled perception and AR support modules are practical in applied settings [6]. Recent advances in deep metric learning for face recognition [7] have significantly improved the performance of face recognition, and augmented reality (AR) has shown promise in healthcare and cognitive support applications [8]. To build on these developments, we propose the combination of real-time face recognition and adaptive AR overlays to offer graded memory support.

This paper presents ReMindAR, an adaptive AR memory assistant for dementia patients. The proposed system uses Multi-task Cascaded Convolutional Networks (MTCNN) for face detection and alignment [9], InceptionResnetV1 pretrained on VGGFace2 for facial embedding extraction [10], the facenet-pytorch implementation for deployment [11], and cosine similarity for identity matching. A Decision Tree classifier then predicts one of four reminder levels from a compact six-feature vector comprising recognition confidence, encounter count, time since last interaction, confidence stability, time of day, and location context. The complete pipeline runs locally on the device to maintain privacy.

The contributions of this paper are: (1) a modular framework for deep face recognition and adaptive reminder prediction, (2) a cognitively-inspired synthetic dataset with a Decision Tree classifier achieving 82.33% accuracy, (3) a real-time progressive web application with AR overlays and speech-to-text logging, and (4) a transparent system design suitable for healthcare applications requiring transparency.

II. Related Work

Recent research has also used photogrammetry to generate realistic 3D objects for AR, improving visualization and practical understanding, which is relevant to the design of intuitive user-facing cues in assistive systems [12].

A. Deep Face Recognition

Deep face recognition is mainly a consequence of embedding-based metric learning methods. The application of triplet loss to embed faces into a discriminative feature space was first introduced in FaceNet, which achieved outstanding performance on LFW [7]. Later, ArcFace and SphereFace introduced angular margin-based losses to further improve discriminability [13], [14].

For practical use, the facenet-pytorch library provides pre-trained models for MTCNN face detection and alignment and InceptionResnetV1 embedding extraction, which are widely used [11]. These models are commonly trained on the VGGFace2 dataset, enabling robust face representation without task-specific retraining [10]. In this work, the pair was selected because it provides a stable end-to-end recognition stack with minimal engineering overhead and direct support for on-device inference.

B. Face Detection and Alignment

Face detection and alignment are critical preprocessing tasks in face recognition systems. MTCNN is a face detector and aligner that adopts a three-stage cascaded network structure, striking a balance between performance and efficiency for real-time processing [9]. Although newer detectors such as RetinaFace can achieve stronger robustness under severe pose, blur, and occlusion [15], MTCNN remains attractive for edge-oriented deployments because it jointly provides bounding boxes and five-point landmarks in a lightweight pipeline. This joint detection-and-alignment capability reduced preprocessing complexity in ReMindAR and supported consistent downstream embedding extraction.

C. Assistive Technologies for Dementia

Lifelogging devices such as SenseCam assist with autobiographical memory recall through retrospective image analysis [16], but do not provide real-time assistance. Other systems, such as COGKNOW [5] and Autominder [17], provided context-aware and adaptive reminder services using environmental sensing and activity recognition, without visual person identification. These findings inform the graded reminder strategy in ReMindAR, where higher intensity levels provide more comprehensive verbal and contextual support.

D. Augmented Reality in Healthcare

Augmented reality has been applied in healthcare for cognitive rehabilitation, navigation assistance, and reminder systems, demonstrating positive impacts on memory and attention in older adults [8]. Wearable AR systems that conduct real-time processing on the device also provide privacy-preserving assistance [18]. However, there is a need for systems that combine face recognition with adaptive information presentation based on recognition confidence and context, which is addressed in ReMindAR.

E. Interpretable Machine Learning

Interpretability is an important factor when considering healthcare models, especially when human intervention and regulatory compliance are a concern. Previous research has indicated that interpretability does not necessarily come at a cost to competitive performance, as shown in [19]. Furthermore, new regulations, such as the EU AI Act, highlight human interpretability and explainability, especially for high-risk medical AI systems [20]. Decision Tree models are applicable in this problem domain, allowing human validation of predictions based on rules.

III. System Architecture

The ReMindAR system is organized as a four-layer architecture, as shown in Fig. 1.

The user interface for camera capturing, AR rendering, face enrollment, and conversation logging is facilitated by the Progressive Web Application. The application layer includes FastAPI REST endpoints for face recognition and data management. It also includes MTCNN Detector [9], InceptionResnetV1 Embedder [10], Cosine Similarity Matcher, and Decision Tree Classifier. Persistent file storage is utilized for storing enrolled face embeddings, conversation history, and model artifacts.

A. Proposed AR Avatar Interaction Module

The proposed AR avatar interaction module acts as a visual interface between the adaptive reminder system and the user [2]. The avatar is designed as a calm, human-like presence in the AR environment and presents reminders through simple visual and audio cues after face recognition and intensity prediction. To minimize cognitive load, the avatar uses limited actions, predictable behavior, and a fixed on-screen position to reduce confusion and distress.

IV. Methodology

A. Face Detection

Face detection is performed using the facenet-pytorch implementation of MTCNN with `keep_all=True` and automatic device selection (`cuda` when available, otherwise `cpu`). MTCNN accepts RGB camera inputs and provides face bounding boxes together with facial landmarks [9]. The three-stage cascade composed of Proposal, Refine, and Output networks suppresses non-face regions and refines final detections with five-point landmark localization. MTCNN was selected because it combines detection and alignment in a single lightweight module, reducing preprocessing complexity and supporting low-latency on-device inference. Although RetinaFace can provide stronger robustness in difficult conditions [15], MTCNN offered a more practical accuracy-efficiency trade-off for the target deployment scenario.

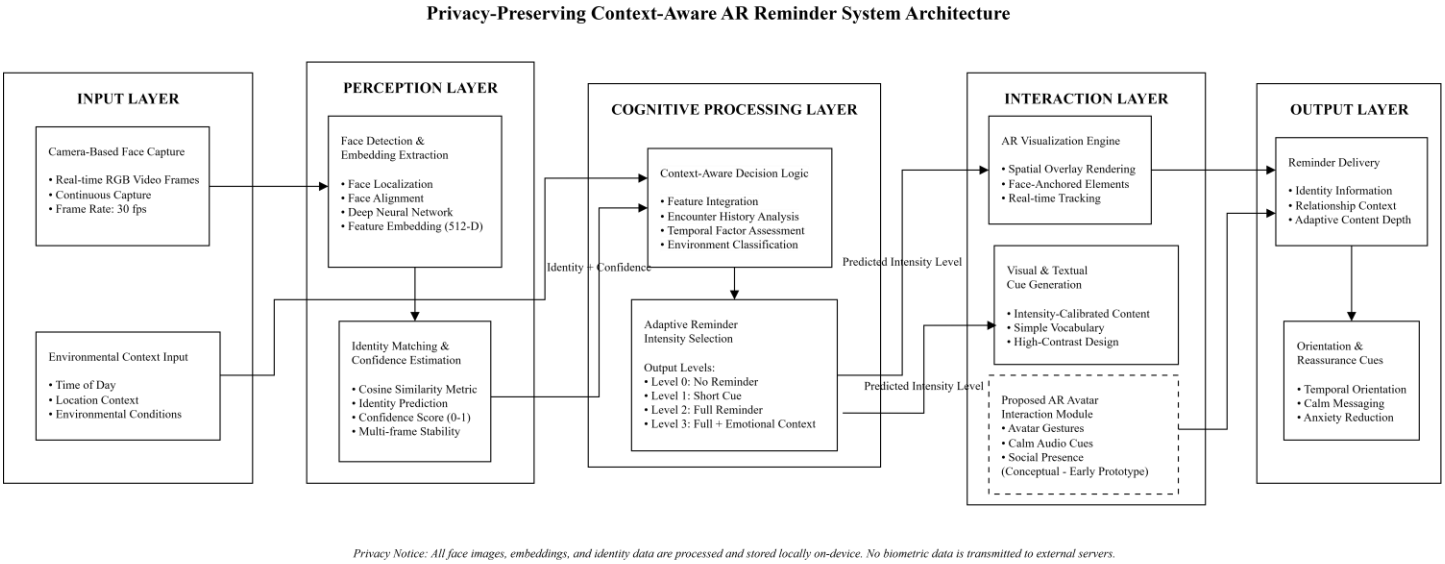


Fig. 1. End-to-end processing pipeline of the ReMindAR system. Camera frames are processed for face detection using MTCNN [9] and embedding extraction using InceptionResnetV1 [11]. Embeddings are matched against enrolled identities using cosine similarity, and contextual features are classified using a Decision Tree classifier for adaptive reminder intensity for AR cue generation.

B. Embedding Extraction

The detected face regions are cropped and resized to 160×160 pixels, and then normalized using the expression $(x-127.5)/128$. The preprocessed images are then fed into the `InceptionResnetV1(pretrained="vggface2")` model in evaluation mode to extract 512-D face embedding vectors. The model combines Inception modules with residual learning [11] and is pre-trained on the VGGFace2 dataset, which contains 3.31 million images of 9,131 individuals with large variations in pose, age, and ethnicity [10]. The InceptionResnetV1 model was selected because it generates high-quality, compact facial embeddings, integrates easily with MTCNN and cosine similarity-based matching in facenet-pytorch, and does not require face-specific retraining for this study. Although more complex ArcFace-style and custom backbones could also be considered, they offered a less favorable trade-off among recognition quality, computational cost, and deployment simplicity for the present system.

C. Face Matching

Identity authentication is performed by computing the cosine similarity between the query embedding vector $\mathbf{q}$ and the reference embedding vector $\mathbf{e}$:

$$sim(q, e) = \frac{q \cdot e}{||q||||e||} \quad (1)$$

The similarity value lies within the range $[-1, 1]$, where a higher value indicates a better match, as defined by the FaceNet metric learning approach [7]. The matcher function returns the identity corresponding to the highest similarity score that exceeds a predefined threshold (default value: 0.70). Scores below this threshold are classified as *unknown*. Following identity matching, the system initiates a request for metadata retrieval and augmented reality (AR) reminder overlay generation.

Figure 2 illustrates the implemented runtime sequence of the deployed API. The API call to the endpoint `/recognize` is designed as a one-shot operation, encompassing face detection, face embedding extraction, cosine similarity matching against `face_db.pkl`, retrieval of conversational context from `conversations_db.pkl`, and Decision Tree-based prediction for reminder intensity used in frontend AR rendering. The runtime sequence is structured such that visual analysis and identity matching are performed first, followed by contextual memory integration, and finally, reminder-level prediction before generating and returning the response to the Progressive Web App (PWA) overlay.

D. Adaptive Reminder Intensity Model

The adaptive reminder intensity model is a central component of ReMindAR because it determines the amount of memory assistance from encounter context rather than issuing a fixed reminder [19]. The task is formulated as a supervised four-class classification problem, where a six-dimensional feature space is mapped to intensity levels ranging from no reminder (Level 0) to full reminder with emotional expressions (Level 3). As depicted in Fig. 3, the confidence scores and contextual variables are evaluated by a trained Decision Tree to determine the reminder intensity.

The input features and cognitive justification are presented in Table I [2].

These six features are determined a priori based on the available variables in the deployed pipeline and cognitive factors documented in the dementia support literature rather than a wrapper-based approach. Features such as recognition confidence are based on the calculation of cosine similarities,

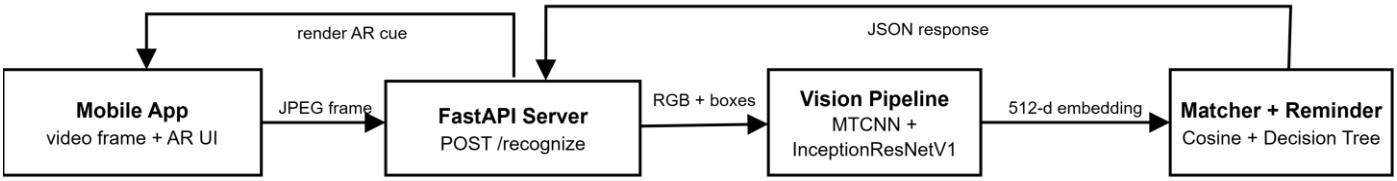


Fig. 2. Runtime recognition and adaptive reminder sequence in ReMindAR. The `/recognize` endpoint integrates visual inference, local identity matching, conversation-context lookup, and reminder-level prediction before returning a structured response to the PWA client.

TABLE I
INPUT FEATURES FOR THE ADAPTIVE REMINDER INTENSITY MODEL

Feature	Type	Rationale
Recognition confidence	Float [0,1]	Higher embedding similarity implies stronger recognition
Encounter count	Integer	Frequent encounters suggest greater familiarity
Time since last	Days	Longer intervals increase memory decay probability
Confidence stability	Float	High variance across frames indicates uncertainty
Time of day	Ordinal	Evening and night hours may impair cognition
Location context	Ordinal	Unfamiliar settings increase cognitive load

while the calculation of the number of encounters and the time elapsed since the last interaction are based on the interaction history. Features like the stability of the recognition confidence are based on the short-term variation of the recognition scores, and the inclusion of the time of day and the location context are based on the environmental effects on cognitive load. Since the feature space is compact and low-dimensional and has a high level of semantic meaning, no dimensionality reduction or principal component analysis is required. Features like the continuous variables are represented numerically, while the ordinal nature of the time of day and the location context is represented as small integer values.

The four reminder levels give a range of assistance, starting with no reminder for high confidence and a positive context at Level 0, followed by a brief reminder at Level 1, a full reminder with a relationship context at Level 2, and a full reminder with an emotionally supportive context at Level 3. As a post-hoc validation of the feature design, the impurity-based feature importance measure for the Decision Tree was used after the training, and recognition confidence was found to be the major predictor, with time since last interaction and location context as strong secondary evidence for the cognitive rationale for the chosen features.

E. Training and Evaluation Setup

In the absence of access to ethically approved real patient data, the reminder model was trained on a synthetic data set consisting of 1,200 encounter instances generated using rule-based cognitive assumptions [2]. The generator used a fixed random seed value of 42 for reproducibility. It sampled *recognition confidence*, *encounter count*, *time since last interaction*, *confidence stability*, *time of day*, and *location context*, and then generated reminder labels according to rule-based cognitive assumptions, where lower recognition confidence, fewer encounters, larger time gaps, evening context, unfamiliar location, and unstable recognition increased reminder intensity up to Level 3.

The generated dataset is imbalanced to match conservative deployment priors: Level 0 = 28 instances (2.3%), Level 1 = 125 instances (10.4%), Level 2 = 224 instances (18.7%), and Level 3 = 823 instances (68.6%). No SMOTE, random oversampling, undersampling, or class weighting was performed. Instead, a 75/25 stratified split was used to preserve the original class balance, yielding 900 training instances and 300 test instances. This preserves the intended operating distribution, but minority-class metrics remain less stable.

The only trainable part of the proposed pipeline is the reminder classifier. MTCNN and InceptionResnetV1 are applied as frozen models in evaluation mode and are therefore not trainable, so parameters such as learning rate, batch size, and epochs are not applicable to the face-recognition component. For reminder prediction, the implementation is based on the CART algorithm with `criterion="gini"`, `max_depth=6`, `min_samples_leaf=20`, and `random_state=42`. These parameters were selected manually rather than through grid search in order to keep the model shallow and easily interpretable for the synthetic data set. The Decision Tree was preferred over alternatives such as SVM and Random Forest because its decision rules are transparent, inference is deterministic, execution is very fast, and the resulting policy can be inspected easily by a clinician. The implementation was tested in a Python environment on Windows 10 and 11, with optional CUDA-supported GPU acceleration for the face-recognition component.

F. Frontend Implementation

The frontend is implemented as a Progressive Web Application (PWA) using HTML, CSS, and JavaScript, enabling mobile installation and offline support through service workers. Camera access is provided through the MediaDevices API, and frames are sent to the backend at fixed intervals. AR overlays are rendered by positioning HTML elements on top of the video stream using the face bounding box coordinates returned by the API. Conversation logging uses the Web Speech API, where speech transcriptions are associated with recognized speakers, extending the SenseCam idea toward

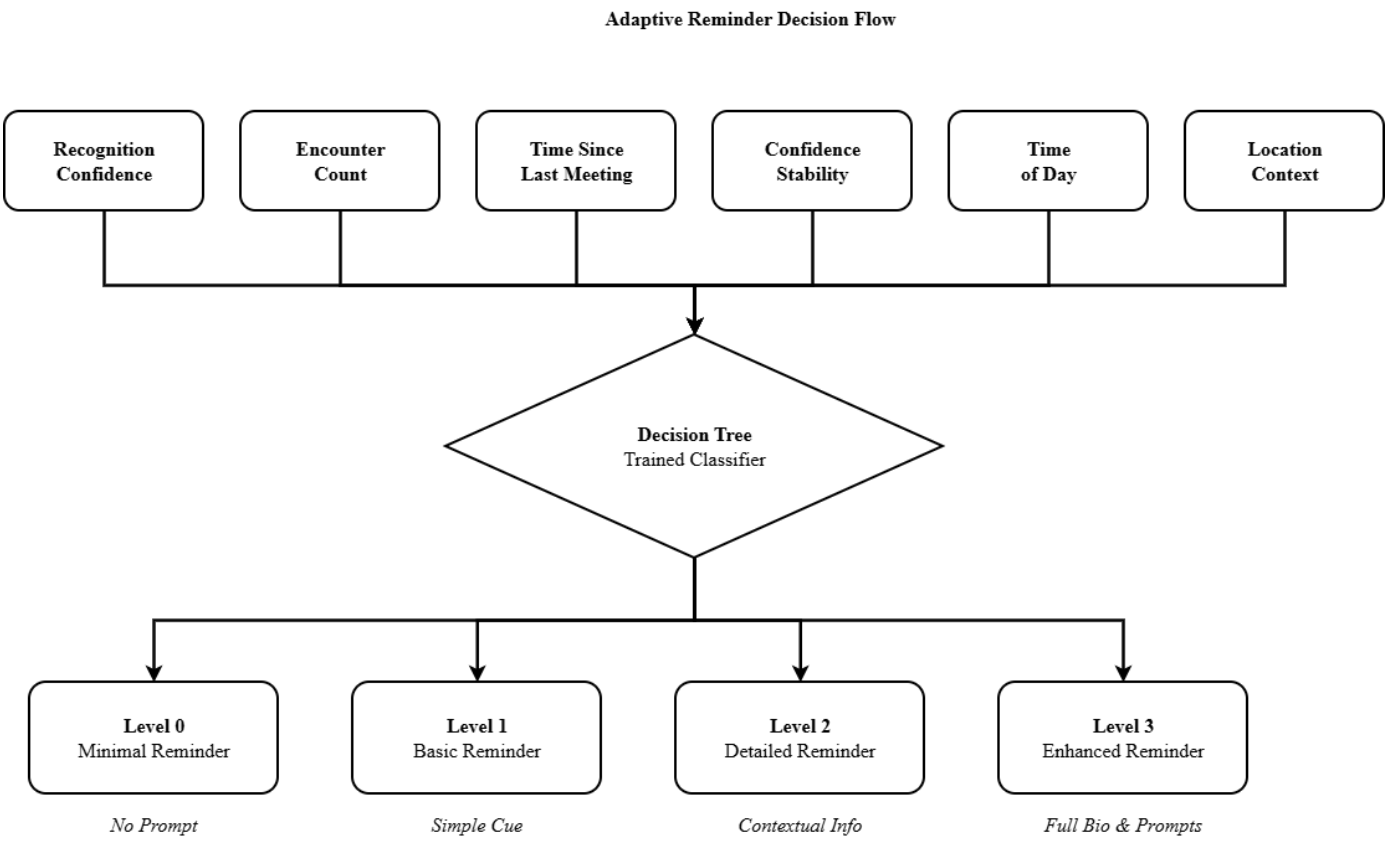


Fig. 3. High-resolution adaptive reminder decision workflow. Runtime encounter features are encoded and passed to the trained Decision Tree classifier, which predicts one of four reminder intensity levels (Level 0–Level 3).

real-time memory support [16]. A runtime screenshot of this interface is shown in Fig. 5.

As depicted in Fig. 4, the process of enrollment and the logging of conversations are part of a closed-loop system. The identity records created via the endpoint /enroll and the conversations created via the endpoint /conversations/name are reused by the /recognize endpoint to gradually adjust the intensity of reminders over time.

G. Proposed AR Avatar Interaction Workflow

The interaction workflow follows dementia care design guidelines [2], [5], [21] and varies reminder communication according to the predicted intensity. At Level 1, the avatar provides a brief visual reminder with short audio output. At Level 2, relationship context is added through on-screen text and audio narration. At Level 3, more comprehensive support is delivered using emotionally informed communication in a calm and supportive tone [22]. Periodic temporal-orientation reminders are also included to support daily routine awareness. The interaction design targets low cognitive load through simple vocabulary, repetition, and a calm visual appearance [5], [21]. Speech output is delivered at a slow rate with pauses, and high-contrast visuals are used to mitigate age-related visual limitations [23]. Recent evidence also shows that technology-based prompts for people with dementia benefit from appropriate prompt modality, task breakdown, and attentional support [24].

TABLE II
CLASSIFICATION PERFORMANCE ON THE TEST SET

Class	Precision	Recall	F1-Score	Support
Level 0: No reminder	0.43	0.86	0.57	7
Level 1: Short cue	0.57	0.74	0.65	31
Level 2: Full reminder	0.62	0.52	0.56	56
Level 3: Full + emotional	0.95	0.92	0.93	206
Weighted average	0.84	0.82	0.83	300

V. RESULTS AND DISCUSSION

This section summarizes the classification performance of the proposed Decision Tree classifier. The trained model achieved an accuracy of 82.33% on a 300-sample test set. The precision, recall, F1-score, and support for each class are listed in Table II.

Table II shows strong performance for the majority class (Level 3), with precision of 0.95 and recall of 0.92, while the minority classes are more difficult because the evaluation set preserves the original class imbalance. In particular, Level 0 is represented by only seven test samples, so its recall of 0.86 is encouraging but statistically unstable, and its precision of

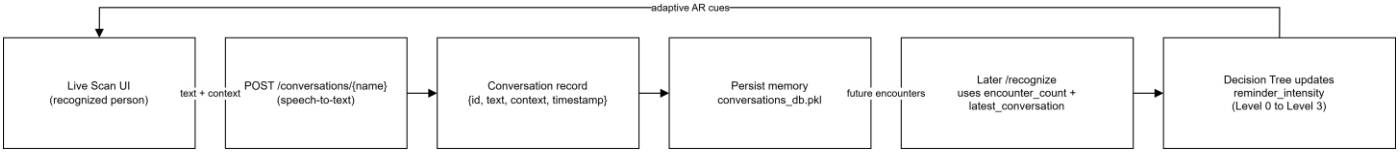


Fig. 4. Enrollment and conversation feedback lifecycle in ReMindAR. Face enrollment persists identity embeddings and metadata, while speech-to-text conversations are stored per person and later reused by the recognition pipeline to update reminder intensity levels.



Fig. 5. PWA interface during live recognition. The screen presents match confidence, person metadata, and reminder-support information over the camera feed.

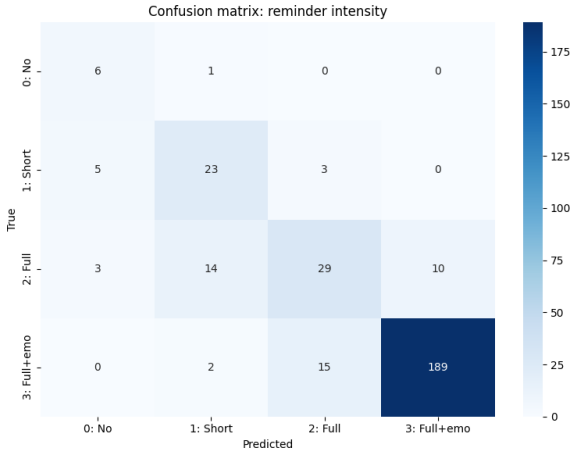


Fig. 6. Confusion matrix for reminder intensity prediction on the 300-sample test set. Most errors occur between adjacent reminder levels.

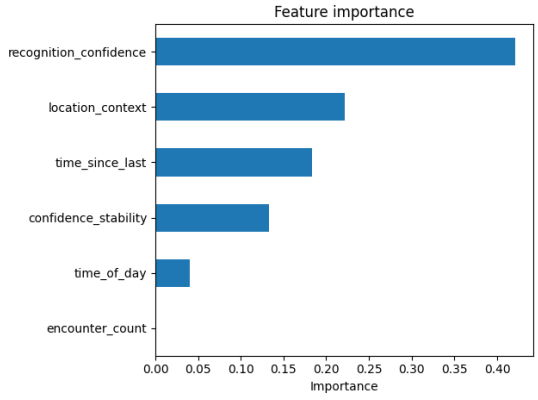


Fig. 7. Relative feature importance for reminder intensity prediction. Recognition confidence is the most influential feature, with location context and time since last encounter contributing strongly. Other features provide complementary context [2].

0.43 reflects a relatively high false-positive rate for the no-reminder class. This behavior is consistent with the underlying label distribution and with the absence of resampling or class weighting during training. Consequently, the weighted average F1-score of 0.83 should be interpreted alongside the per-class metrics, since aggregate scores are dominated by the frequent Level 3 cases. Most errors occur between adjacent reminder levels, which is clinically less problematic than confusing Level 0 and Level 3.

Figure 6 confirms that errors are concentrated between nearby reminder levels rather than between extreme levels. Figure 7 reports impurity-based (Gini) feature importance from the trained Decision Tree. Recognition confidence is the strongest predictor, while time since last interaction and location context also contribute substantially, consistent with cognitive theories of familiarity, uncertainty, and memory decay [2]. The low-dimensional feature set therefore appears sufficient for interpretable reminder prediction without additional feature compression.

Face detection and embedding extraction take approximately 50 ms per frame with GPU acceleration, while Decision Tree inference takes less than one millisecond, supporting smooth AR overlay rendering.

VI. Limitations

However, one of the major limitations of this study is that the training data is synthetic rather than being derived directly from patient interactions. Although the rules are based on cognitive assumptions supported by literature on dementia care, the labels derived are idealized and need to be validated with patients and caregivers.

Class imbalance is another limitation, particularly for Level 0 instances, which constitute only 2.3% of the dataset. Because no resampling or cost-sensitive learning was applied, minority-class estimates remain sensitive to small changes in sample count. This leads to lower precision for no-reminder predictions and, at times, unnecessary reminders, which is arguably a safer failure mode than withholding support when it is needed. The single-camera setup may also degrade performance under occlusion or extreme head pose, even though MTCNN and RetinaFace have demonstrated robustness in such conditions. Finally, differences across browsers and languages may affect the quality of speech-to-text transcription obtained from the Web Speech API.

Lastly, the AR avatar module remains at the conceptual stage of the current implementation. Future studies should include user evaluations to assess the avatar design, speech characteristics, and psychological impact.

VII. Conclusion and Future Work

In this study, we proposed ReMindAR, an adaptive AR-based memory aid system for dementia patients that combines deep face recognition with context-aware reminder intensity prediction. The system integrates MTCNN for face detection and alignment, InceptionResnetV1 embeddings, cosine-similarity matching, and an interpretable Decision Tree classifier in a privacy-preserving on-device pipeline. The reminder model achieved 82.33% accuracy on four-class intensity prediction while retaining the transparency required for clinical interpretation.

Future work will include clinical validation with dementia patients and caregivers, expansion of the dataset through ethically approved real-world interactions, and systematic comparison with alternative reminder classifiers such as cost-sensitive trees, Random Forests, and margin-based methods. Additional work will also investigate wearable AR devices for hands-free operation and assess whether adaptive reminders can reduce social disengagement in dementia care.

References

[1] World Health Organization, “Global status report on the public health response to dementia,” WHO, Geneva, Switzerland, 2021. [Online]. Available: https://www.who.int/publications/i/item/9789240033245

[2] F. Meiland *et al.*, “Technologies to support community-dwelling persons with dementia: A position paper on issues regarding development, usability, effectiveness, and cost-effectiveness, deployment, and ethics,” *JMIR Rehabil. Assist. Technol.*, vol. 4, no. 1, p. e1, Jan. 2017.

[3] Kaviyaraj, R., and M. Uma. ”A survey on future of augmented reality with AI in education.” 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS). IEEE, 2021.

[4] Kaviyaraj, R. ”Generative Artificial Intelligence: Transforming the Future.” 2024 International Conference on Emerging Technologies and Innovation for Sustainability (EmergIN). IEEE, 2024.

[5] R. Orpwood, C. Gibbs, T. Adlam, R. Faulkner, and D. Meegahawatte, “The design of smart homes for people with dementia—user interface aspects,” *Univ. Access Inf. Soc.*, vol. 4, no. 2, pp. 156–164, Dec. 2005.

[6] Kaviyaraj, R., and M. Uma. ”An augmented reality framework for education: deep learning integration and impact evaluation.” IEEE Access (2025).

[7] F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A unified embedding for face recognition and clustering,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Boston, MA, USA, Jun. 2015, pp. 815–823.

[8] M. Schall, S. Schuck, and M. Thust, “Augmented reality for cognitive rehabilitation: A systematic literature review,” *J. Rehabil. Assist. Technol.*, vol. 9, pp. 1–15, 2021.

[9] K. Zhang, Z. Zhang, Z. Li, and Y. Qiao, “Joint face detection and alignment using multitask cascaded convolutional networks,” *IEEE Signal Process. Lett.*, vol. 23, no. 10, pp. 1499–1503, Oct. 2016.

[10] Q. Cao, L. Shen, W. Xie, O. M. Parkhi, and A. Zisserman, “VGGFace2: A dataset for recognising faces across pose and age,” in *Proc. IEEE Int. Conf. Autom. Face Gesture Recognit. (FG)*, Xi’an, China, May 2018, pp. 67–74.

[11] T. Esler, “facenet-pytorch: Pretrained PyTorch face detection and recognition models,” GitHub, 2020. [Online]. Available: https://github.com/timesler/facenet-pytorch

[12] Ravichandran, Kaviyaraj, and Uma Mohan. ”Transforming Education with Photogrammetry: Creating Realistic 3D Objects for Augmented Reality Applications.” Computer Modeling in Engineering & Sciences 142.1 (2025): 185.

[13] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive angular margin loss for deep face recognition,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Sep. 2022, pp. 4690–4699.

[14] W. Liu, Y. Wen, Z. Yu, M. Li, B. Raj, and L. Song, “SphereFace: Deep hypersphere embedding for face recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jan. 2018, pp. 212–220.

[15] J. Deng, J. Guo, E. Ververas, I. Kotsia, and S. Zafeiriou, “RetinaFace: Single-shot multi-level face localisation in the wild,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Seattle, WA, USA, Jun. 2020, pp. 5203–5212.

[16] S. Hodges *et al.*, “SenseCam: A retrospective memory aid,” in *Proc. 8th Int. Conf. Ubiquitous Comput. (UbiComp)*, Orange County, CA, USA, Sep. 2006, pp. 177–193.

[17] M. E. Pollack *et al.*, “Autominder: An intelligent cognitive orthotic system for people with memory impairment,” *Robot. Auton. Syst.*, vol. 44, no. 3–4, pp. 273–282, Sep. 2003.

[18] J. T. Matthews, J. H. Lingler, G. B. Campbell, A. E. Hunsaker, L. Hu, B. R. Pires, M. Hebert, and R. Schulz, “Usability of a wearable camera system for dementia family caregivers,” *J. Healthc. Eng.*, vol. 6, no. 2, pp. 213–238, 2015.

[19] R. Caruana *et al.*, “Intelligible models for healthcare: Predicting pneumonia risk and hospital 30-day readmission,” in *Proc. 21st ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, Sydney, Australia, Aug. 2015, pp. 1721–1730.

[20] European Commission, “Proposal for a regulation laying down harmonised rules on artificial intelligence (Artificial Intelligence Act),” Brussels, Belgium, Apr. 2021. [Online]. Available: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:52021PC0206

[21] A. Astell *et al.*, “Technology and dementia: The future is now,” *Dementia Geriatr. Cogn. Disord.*, vol. 47, no. 3, pp. 131–139, 2019.

[22] T. W. Bickmore and R. W. Picard, “Establishing and maintaining long-term human-computer relationships,” *ACM Trans. Comput.-Hum. Interact.*, vol. 12, no. 2, pp. 293–327, Jun. 2005.

[23] J. Buchner, K. Buntins, and M. Kerres, “The impact of augmented reality on cognitive load and performance: A systematic review,” *J. Comput. Assist. Learn.*, vol. 38, no. 1, pp. 285–303, Feb. 2022.

[24] M. Cannings, R. Brookman, S. Parker, L. Hoon, A. Ono, H. Kawata, H. Matsukawa, and C. B. Harris, “Optimizing technology-based prompts for supporting people living with dementia in completing activities of daily living at home: Experimental approach to prompt modality, task breakdown, and attentional support,” *JMIR Aging*, vol. 7, p. e56055, Aug. 2024, doi: 10.2196/56055.